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End cover assembly, energy storage apparatus, and electricity-consumption device

Abstract

An end cover assembly, an energy storage apparatus, and an electricity-consumption device are provided in the disclosure to improve a production yield. The end cover assembly includes a top cover, an upper plastic member, a conductive block, a terminal post, and a sealing member. A mounting hole of the top cover extends through the top cover in a thickness direction of the top cover. The upper plastic member is mounted to the top cover. A through hole of the upper plastic member extends through the upper plastic member in the thickness direction of the upper plastic member and is in communication with the mounting hole. The conductive block is mounted to the upper plastic member. A fitting hole of the conductive block extends through the conductive block in a thickness direction of the conductive block and is in communication with the through hole.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority under 35 U.S.C. § 119(a) to Chinese Patent Application No. 202310091034.5, filed Feb. 9, 2023, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The disclosure relates to the field of energy storage technologies, and in particular, to an end cover assembly, an energy storage apparatus, and an electricity-consumption device.

BACKGROUND

(3) In existing energy storage apparatuses (for example, secondary batteries), a sealing member, an upper plastic member, and a lower plastic member are generally arranged between a terminal post and a top cover to achieve sealing. However, the terminal post and a conductive block will be subjected to relatively large external forces during riveting, and the sealing member easily loosens, thereby reducing the sealing performance between the terminal post and the top cover, and lowering a production yield of the energy storage apparatuses.

SUMMARY

(4) According to a first aspect, an end cover assembly applicable to a battery cell is provided in the disclosure. The end cover assembly includes a top cover, an upper plastic member, a conductive block, a terminal post, and a sealing member. The top cover defines a mounting hole. The mounting hole extends through the top cover in a thickness direction of the top cover. The upper plastic member is mounted on the top cover. The upper plastic member defines a through hole extending through the upper plastic member in a thickness direction of the upper plastic member and being in communication with the mounting hole. The conductive block is mounted on the upper plastic member. The conductive block defines a fitting hole. The fitting hole extends through the conductive block in a thickness direction of the conductive block and is in communication with the through hole. The conductive block includes a protruding ring. The protruding ring is disposed on the conductive block and surrounds an inner wall of the fitting hole. The protruding ring abuts against the upper plastic member. The terminal post includes a flange portion and a post portion. The post portion is disposed on a top side of the flange portion. The post portion extends through the mounting hole, the through hole, and the fitting hole and is fixedly connected to the conductive block. The sealing member is sleeved on the post portion and clamped between the post portion and a wall of the mounting hole. A top surface of the sealing member abuts against a bottom surface of the upper plastic member, and a bottom surface of the sealing member abuts against a top surface of the flange portion.

(5) According to a second aspect, an energy storage apparatus is provided in the disclosure. The energy storage apparatus includes a housing and the end cover assembly provided in any one of the above embodiments, the end cover assembly is mounted on a top side of the housing.

(6) According to a third aspect, an electricity-consumption device is provided in the disclosure. The electricity-consumption device includes the above-mentioned energy storage apparatus, and the energy storage apparatus is configured to power the electricity-consumption device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To describe the technical solutions in the embodiments of the disclosure more clearly, the following describes the accompanying drawings for describing embodiments of the disclosure.

(2) FIG. **1** is a schematic structural view of an energy storage apparatus according to embodiments of the disclosure.

(3) FIG. **2** is a schematic structural view of an end cover assembly of the energy storage apparatus illustrated in FIG. **1**.

(4) FIG. **3** is a schematic exploded structural view of the end cover assembly illustrated in FIG. **2**.

(5) FIG. **4** is a schematic cross-sectional structural view of the end cover assembly illustrated in FIG. **2**, taken along line I-I.

(6) FIG. **5** is a schematic structural view illustrating a lower plastic member, a top cover, an explosion-proof valve, a positive-electrode deformation member, and a negative-electrode deformation member of the end cover assembly illustrated in FIG. **3**.

(7) FIG. **6** is a schematic exploded structural view of a positive-electrode unit of the end cover assembly illustrated in FIG. **3**.

(8) FIG. **7** is a schematic cross-sectional structural view of a first upper plastic member of the positive-electrode unit illustrated in FIG. **6**, taken along line II-II.

(9) FIG. **8** is a schematic structural view of a first conductive block of the positive-electrode unit illustrated in FIG. **6**, viewed from another direction.

(10) FIG. **9** is a schematic exploded structural view of a negative-electrode unit of the end cover assembly illustrated in FIG. **3**.

(11) FIG. **10** is a schematic diagram of an electricity-consumption device according to embodiments of the disclosure.

(12) FIG. **11** is an enlarged schematic structural view of Circle A in FIG. **4**.

(13) FIG. **12** is an enlarged schematic structural view of Circle B in FIG. **4**.

(14) Reference numbers in the accompanying drawings are described as follows: energy storage apparatus **100**, housing **110**, end cover assembly **120**, lower plastic member **10**, top cover **20**, explosion-proof valve **30**, positive-electrode deformation member **40**, negative-electrode deformation member **50**, positive-electrode unit **60**, negative-electrode unit **70**, explosion-proof grid **11**, first electrolyte injection hole **101**, first avoidance groove **102**, second avoidance groove **103**, third mounting hole **104**, fourth mounting hole **105**, explosion-proof hole **201**, second electrolyte injection hole **202**, first mating hole **203**, second mating hole **204**, first mounting hole **205**, second mounting hole **206**, first upper plastic member **61**, first conductive block **62**, first terminal post **63**, first sealing member **64**, first connector **65**, first mounting portion **611**, first mating portion **612**, first fitting groove **613**, first avoidance hole **614**, first identification through-hole **615**, first groove top surface **616**, first groove bottom surface **617**, identification portion **611*a***, first mounting groove **618**, first through hole **619**, first extension portion **621**, first welding portion **622**, first top surface **623**, first bottom surface **624**, first protruding ring **622*a***, second top surface **625**, second bottom surface **626**, first fitting hole **627**, first fitting portion **628**, first connecting portion **629**, first flange portion **631**, first post portion **632**, first sealing portion **641**, first secondary sealing portion **642**, second upper plastic member **71**, second conductive block **72**, second terminal post **73**, second sealing member **74**, second connector **75**, second mounting portion **711**, second mating portion **712**, second fitting groove **713**, second avoidance hole **714**, second identification through-hole **715**, second groove top surface **716**, second groove bottom surface **717**, second mounting groove **718**, second through hole **719**, second extension portion **721**, second welding portion **722**, third top surface **723**, third bottom surface **724**, second protruding ring **722*a***, fourth top surface **725**, fourth bottom surface **726**, second fitting hole **727**, second fitting portion **728**, second connecting portion **729**, second flange portion **731**, second post portion **732**, second sealing portion **741**, second secondary sealing portion **742**, electricity-consumption device **1**.

DETAILED DESCRIPTION

(15) The following clearly describes the technical solutions in the embodiments of the disclosure with reference to the accompanying drawings in the embodiments of the disclosure.

(16) An end cover assembly, an energy storage apparatus, and an electricity-consumption device are provided in the disclosure to achieve an improved production yield.

(17) According to a first aspect, an end cover assembly applicable to a battery cell is provided in the disclosure. The end cover assembly includes a top cover, an upper plastic member, a conductive block, a terminal post, and a sealing member. The top cover defines a mounting hole. The mounting hole extends through the top cover in a thickness direction of the top cover. The upper plastic member is mounted on the top cover. The upper plastic member defines a through hole extending through the upper plastic member in a thickness direction of the upper plastic member and being in communication with the mounting hole. The conductive block is mounted on the upper plastic member. The conductive block defines a fitting hole. The fitting hole extends through the conductive block in a thickness direction of the conductive block and is in communication with the through hole. The conductive block includes a protruding ring. The protruding ring is disposed on the conductive block and surrounds an inner wall of the fitting hole. The protruding ring abuts against the upper plastic member after assembly. The terminal post includes a flange portion and a post portion. The post portion is disposed on a top side of the flange portion. The post portion extends through the mounting hole, the through hole, and the fitting hole and is fixedly connected to the conductive block. The sealing member is sleeved on the post portion and clamped between the post portion and a wall of the mounting hole. A top surface of the sealing member abuts against a bottom surface of the upper plastic member, and a bottom surface of the sealing member abuts against a top surface of the flange portion.

(18) In embodiments of the disclosure, a ratio of a thickness of the protruding ring to a height of the protruding ring ranges from 0.2 to 0.8.

(19) In embodiments of the disclosure, a height of the protruding ring ranges from 0.1 mm to 1.2 mm.

(20) In embodiments of the disclosure, a thickness of the protruding ring ranges from 0.2 mm to 2.4 mm.

(21) In embodiments of the disclosure, the sealing member includes a sealing portion and a secondary sealing portion. The secondary sealing portion is positioned on a bottom of the sealing portion and surrounds the sealing portion. The sealing portion is sleeved on the post portion and clamped between the post portion and the wall of the mounting hole of the top cover.

(22) A surface of the sealing portion that faces away from the secondary sealing portion abuts against the upper plastic member. A surface of the secondary sealing portion that faces away from the sealing portion abuts against the top surface of the flange portion.

(23) In embodiments of the disclosure, the conductive block includes a fitting portion and a connecting portion. The connecting portion is disposed on a top side of the fitting portion and extends outwards relative to a peripheral surface of the fitting portion to form the protruding ring. The upper plastic member covers the peripheral surface of the fitting portion.

(24) In embodiments of the disclosure, the upper plastic member defines a mounting groove, and the fitting portion is received in the mounting groove.

(25) In embodiments of the disclosure, the through hole is in communication with the mounting groove, the fitting hole extends through the fitting portion and the connecting portion, and the protruding ring is disposed on a surface of the fitting portion away from the connecting portion.

(26) In embodiments of the disclosure, the upper plastic member includes a mounting portion and a mating portion fixedly connected to one side of the mounting portion. The mounting portion defines a fitting groove and an identification through-hole. An opening of the fitting groove is defined on a surface of the mounting portion facing towards the mating portion. The fitting groove is in communication with the mounting groove. An opening of the identification through-hole is

defined on a top surface of the mounting portion, and the identification through-hole is in communication with the fitting groove. The mating portion defines the mounting groove and the through hole. The conductive block includes an extension portion and a welding portion. The welding portion is fixedly connected to one side of the extension portion. The extension portion is mounted to the fitting groove. A top surface of the extension portion covers the identification through-hole. The extension portion is different from the mounting portion in color. The welding portion includes the fitting portion and the connecting portion. The welding portion is received in the mounting groove.
(27) In embodiments of the disclosure, the fitting groove includes a groove top surface and a groove bottom surface opposite the groove top surface. The top surface of the extension portion abuts against the groove top surface. A bottom surface of the extension portion abuts against the groove bottom surface.
(28) In embodiments of the disclosure, a top surface of the welding portion protrudes relative to the top surface of the mounting portion.
(29) In embodiments of the disclosure, a surface of the mounting portion facing towards the mating portion is spaced from an outer peripheral surface of the welding portion.
(30) In embodiments of the disclosure, the end cover assembly includes a positive-electrode unit and a negative-electrode unit. The positive-electrode unit and the negative-electrode unit are both mounted to the top cover and are arranged at intervals. The negative-electrode unit includes one upper plastic member, one conductive block, one terminal post, and one sealing member. The positive-electrode unit includes one upper plastic member, one conductive block, one terminal post, and one sealing member.
(31) According to a second aspect, an energy storage apparatus is provided in the disclosure. The energy storage apparatus includes a housing and the end cover assembly provided in any one of the above embodiments, the end cover assembly is mounted on a top side of the housing.
(32) According to a third aspect, an electricity-consumption device **1** is provided in the disclosure. The electricity-consumption device **1** includes the above-mentioned energy storage apparatus, and the energy storage apparatus is configured to power the electricity-consumption device **1**.
(33) FIG. **1** is a schematic structural view of an energy storage apparatus **100** according to the embodiments of the disclosure. For ease of illustration, a length direction of the energy storage apparatus **100** is defined as an X-axis direction, a width direction of the energy storage apparatus **100** is defined as a Y-axis direction, a height direction of the energy storage apparatus **100** is defined as a Z-axis direction. The X-axis direction, the Y-axis direction, and the Z-axis direction are perpendicular to each other.
(34) The energy storage apparatus **100** includes a housing **110**, a jelly-roll (not illustrated), and an end cover assembly **120**. The housing **110** has an opening (not illustrated) and an accommodation cavity (not illustrated). The jelly-roll is accommodated in the accommodation cavity. The accommodation cavity is also configured for accommodating an electrolyte, and the jelly-roll is soaked in the electrolyte. The end cover assembly **120** is mounted on a top side of the housing **110** and seals the opening. Exemplarily, the energy storage apparatus **100** may be a prismatic battery. In other embodiments, the energy storage apparatus **100** may also be a cylindrical battery or other batteries.
(35) It may be noted that, directional terms such as “top”, “bottom”, “left”, and “right” in the disclosure are described with reference to orientations illustrated in FIG. **1**. “Top” is defined as facing a positive direction of the Z-axis. “Bottom” is defined as facing a negative direction of the Z-axis. “Right” is defined as facing a positive direction of the X-axis. “Left” is defined as facing a negative direction of the X-axis. The above terms do not explicitly or implicitly indicate that apparatuses/devices or components referred to must have a certain orientation, be configured or operated in a certain orientation, and therefore are not limitations on the disclosure.
(36) As illustrated in FIGS. **2** to **4**, FIG. **2** is a schematic structural view of the end cover assembly

120 of the energy storage apparatus **100** illustrated in FIG. **1**, FIG. **3** is a schematic exploded structural view of the end cover assembly **120** illustrated in FIG. **2**, and FIG. **4** is a schematic cross-sectional structural view of the end cover assembly **120** illustrated in FIG. **2**, taken along line I-I. "Taken along line I-I" refers to cut along a plane where line I-I is positioned. Similar illustrations hereinafter may be interpreted in the same manner.

(37) The end cover assembly **120** includes a lower plastic member **10**, a top cover **20**, an explosion-proof valve **30**, a positive-electrode deformation member **40**, a negative-electrode deformation member **50**, a positive-electrode unit **60**, and a negative-electrode unit **70**. The top cover **20** is mounted on a top side of the lower plastic member **10**. The explosion-proof valve **30**, the positive-electrode deformation member **40**, and the negative-electrode deformation member **50** are all mounted to the top cover **20**. The positive-electrode deformation member **40** and the negative-electrode deformation member **50** are positioned at two opposite sides of the explosion-proof valve **30**, respectively. The positive-electrode deformation member **40** and the negative-electrode deformation member **50** are both spaced apart from the explosion-proof valve **30**. The positive-electrode unit **60** and the negative-electrode unit **70** are both mounted to the top cover **20**, and are positioned at opposite sides of the explosion-proof valve **30**, respectively. The positive-electrode unit **60** and the positive-electrode deformation member **40** are positioned at the same side of the explosion-proof valve **30**. The positive-electrode unit **60** covers the positive-electrode deformation member **40**. The negative-electrode unit **70** and the negative-electrode deformation member **50** are positioned at the same side of the explosion-proof valve **30**. The negative-electrode unit **70** covers the negative-electrode deformation member **50**.

(38) Both the positive-electrode deformation member **40** and the negative-electrode deformation member **50** are configured to deform in response to an increase in a pressure inside the energy storage apparatus **100**. When a gas pressure inside the energy storage apparatus **100** exceeds a preset pressure threshold value, the positive-electrode deformation member **40** can deform to be in contact with a conductive block of the positive-electrode unit **60** so that the positive-electrode unit **60** is short-circuited externally, and the negative-electrode deformation member **50** can deform to be in contact with a conductive block of the negative-electrode unit **70** so that the negative-electrode unit **70** is short-circuited externally. As a result, a large short-circuit current is generated to cause a phenomenon of fusing and top-cutting at the positive-electrode deformation member **40** and a bottom of the conductive block of the positive-electrode unit **60** so that the positive-electrode deformation member **40** is open-circuited with the conductive block of the positive-electrode unit **60**; and a large short-circuit current is generated to cause a phenomenon of fusing and top-cutting at the negative-electrode deformation member **50** and a bottom of the conductive block of the negative-electrode unit **70** so that the negative-electrode deformation member **50** is open-circuited with the conductive block of the negative-electrode unit **70**. This can avoid an overcharging and explosion of the energy storage apparatus **100**, ensuring the safety and reliability of the energy storage apparatus **100**.

(39) FIG. **5** is a schematic structural view illustrating the lower plastic member **10**, the top cover **20**, the explosion-proof valve **30**, the positive-electrode deformation member **40**, and the negative-electrode deformation member **50** of the end cover assembly **120** illustrated in FIG. **3**.

(40) The lower plastic member **10** includes an explosion-proof grid **11**. The explosion-proof grid **11** extends through a top surface (no reference number is illustrated) of the lower plastic member **10** and a bottom surface (no reference number is illustrated) of the lower plastic member **10**. The explosion-proof grid **11** is positioned in the middle of the lower plastic member **10**. The lower plastic member **10** defines a first electrolyte injection hole **101**, a first avoidance groove **102**, a second avoidance groove **103**, a third mounting hole **104**, and a fourth mounting hole **105**. The first electrolyte injection hole **101** extends through the lower plastic member **10** in a thickness direction of the lower plastic member **10** (i.e., the Z-axis direction as illustrated). In an embodiment, the first electrolyte injection hole **101** is positioned in the middle of the lower plastic member **10**, is

positioned at the right side of the explosion-proof grid **11**, and is spaced apart from the explosion-proof grid **11**. Exemplarily, the first electrolyte injection hole **101** is a circular hole. In other embodiments, the first electrolyte injection hole **101** may also be a square hole or other irregular holes.

(41) In the X-axis direction, the first avoidance groove **102** and the second avoidance groove **103** are defined at two opposite sides of the explosion-proof grid **11**, respectively. The first avoidance groove **102** and the second avoidance groove **103** are both spaced apart from the explosion-proof grid **11**. The first avoidance groove **102** is symmetrically defined with the second avoidance groove **103** about the explosion-proof grid **11**. In an embodiment, the first avoidance groove **102** is defined at the right side of the explosion-proof grid **11**, is positioned at a side of the first electrolyte injection hole **101** facing away from the explosion-proof grid **11**, and is spaced apart from the first electrolyte injection hole **101**. The second avoidance groove **103** is defined at the left side of the first electrolyte injection hole **101**. Both an opening of the first avoidance groove **102** and an opening of the second avoidance groove **103** are defined on the top surface (no reference number is illustrated) of the lower plastic member **10**. Both the first avoidance groove **102** and the second avoidance groove **103** are recessed from the top surface of the lower plastic member **10** towards the bottom surface (no reference number is illustrated) of the lower plastic member **10**, that is, recessed in the negative direction of the Z-axis as illustrated). Exemplarily, both the first avoidance groove **102** and the second avoidance groove **103** are circular grooves. In other embodiments, the first avoidance groove **102** and the second avoidance groove **103** may also be square grooves or other irregular grooves.

(42) The third mounting hole **104** is defined at a side of the first avoidance groove **102** facing away from the first electrolyte injection hole **101**, and is spaced apart from the first avoidance groove **102**. The fourth mounting hole **105** is defined at a side of the second avoidance groove **103** facing away from the explosion-proof grid **11**, and is spaced apart from the second avoidance groove **103**. The third mounting hole **104** is symmetrically defined with the fourth mounting hole **105** about the explosion-proof grid **11**. In an embodiment, both the third mounting hole **104** and the fourth mounting hole **105** extend through the lower plastic member **10** in the thickness direction of the lower plastic member **10**. Exemplarily, both the third mounting hole **104** and the fourth mounting hole **105** are circular holes, and in other embodiments, the third mounting hole **104** and the fourth mounting hole **105** may also be square holes or other irregular holes.

(43) As illustrated in FIGS. **4** and **5**, the top cover **20** defines an explosion-proof hole **201**, a second electrolyte injection hole **202**, a fitting hole, and a mounting hole. In this embodiment, the explosion-proof hole **201** extends through the top cover **20** in the thickness direction of the top cover **20** (i.e., the Z-axis direction as illustrated). In an embodiment, the explosion-proof holes **201** are defined in the middle of the top cover **20**. The explosion-proof hole **201** is in communication with the inside of the energy storage apparatus **100** through the explosion-proof grid **11**. Exemplarily, the explosion-proof hole **201** is an oval hole, and the second electrolyte injection hole **202** is a circular hole. In other embodiments, the explosion-proof hole **201** may be a circular hole, a square hole, or other shaped holes.

(44) It may be noted that, terms such as “inside/interior” and “outside/exterior” mentioned herein for describing the end cover assembly **120** in the embodiments of the disclosure are described with reference to the orientations of the energy storage apparatus **100** illustrated in FIG. **1**. The term “outside/exterior” refers to a side facing away from the housing **110**, and the term “inside/interior” refers to a side facing toward the housing **110**. Similar illustrations hereinafter may be interpreted in the same manner.

(45) The second electrolyte injection hole **202** is defined in the middle of the top cover **20**, is positioned at the right side of the explosion-proof hole **201**, and is spaced apart from the explosion-proof hole **201**. In an embodiment, the second electrolyte injection hole **202** extends through the top cover **20** in the thickness direction of the top cover **20**. The second electrolyte injection hole

202 is in communication with the first electrolyte injection hole **101** of the lower plastic member **10**. The electrolyte may be sequentially injected into the accommodation cavity of the housing **110** (as illustrated in FIG. **1**) through the second electrolyte injection hole **202** of the top cover **20** and the first electrolyte injection hole **101** of the lower plastic member **10**, thereby achieving injection of the electrolyte into the energy storage apparatus **100**. Exemplarily, the second electrolyte injection hole **202** is a circular hole. In other embodiments, the second electrolyte injection hole **202** may be a square hole or other irregular holes.
(46) The fitting hole is defined at an edge of the top cover **20** and is spaced apart from the second electrolyte injection hole **202**. In an embodiment, the fitting hole extends through the top cover **20** in the thickness direction of the top cover **20**. In the embodiments, two fitting holes are defined and include a first mating hole **203** and a second mating hole **204**. In the X-axis direction, the first mating hole **203** and the second mating hole **204** are defined at two opposite sides of the explosion-proof hole **201**, respectively. The first mating hole **203** is symmetrically defined with the second mating hole **204** about the explosion-proof hole **201**. In an embodiment, the first mating hole **203** is defined at the right side of the explosion-proof hole **201**, is positioned at a side of the second electrolyte injection hole **202** facing away from the explosion-proof hole **201**, and is spaced apart from the explosion-proof hole **201**. The second mating hole **204** is defined at the left side of the first electrolyte injection hole **101**. The first mating hole **203** is in communication with the first avoidance groove **102**. The second mating hole **204** is in communication with the second avoidance groove **103**. Exemplarily, both the first mating hole **203** and the second mating hole **204** are circular holes. In other embodiments, the first mating hole **203** and the second mating hole **204** may also be square holes or other irregular holes.
(47) The mounting hole is positioned at the edge of the top cover **20** and is spaced apart from the fitting hole. In an embodiment, the mounting hole extends through the top cover **20** in the thickness direction of the top cover **20**. In the embodiments, two mounting holes are defined and include a first mounting hole **205** and a second mounting hole **206**. In the embodiments of the disclosure, the mounting hole may be implemented as at least one of the first mounting hole **205** or the second mounting hole **206**. In an embodiment, the first mounting hole **205** is defined at a side of the first mating hole **203** facing away from the second electrolyte injection hole **202**, and is spaced apart from the first mating hole **203**. The second mounting hole **206** is defined at a side of the second mating hole **204** facing away from the explosion-proof hole **201**, and is spaced apart from the second mating hole **204**. The first mounting hole **205** is symmetrically defined with the second mounting hole **206** about the explosion-proof hole **201**. In addition, the first mounting hole **205** is in communication with the third mounting hole **104** of the lower plastic member **10**, and the second mounting hole **206** is in communication with the fourth mounting hole **105** of the lower plastic member **10**. Exemplarily, both the first mounting hole **205** and the second mounting hole **206** are circular holes. In some other embodiments, the first mounting hole **205** and the second mounting hole **206** may also be square holes or other irregular holes.
(48) The explosion-proof valve **30** is mounted in the explosion-proof hole **201** and is fixedly connected to a wall of the explosion-proof hole **201**. Exemplarily, the explosion-proof valve **30** may be fixedly connected to the wall of the explosion-proof hole **201** by welding, so that the explosion-proof valve **30** is mounted in the explosion-proof hole **201**. It may be understood that, the explosion-proof hole **201** is in communication with the inside and the outside of the energy storage apparatus **100**; thus when a gas pressure inside the energy storage apparatus **100** is excessively high, the explosion-proof valve **30** will be broken under the effect of the gas pressure, so that gas inside the energy storage apparatus **100** can sequentially pass through the explosion-proof hole **201** and the explosion-proof grid **11** of the lower plastic member **10** and be discharged to the outside of the energy storage apparatus **100** in real time. Therefore, explosion of the energy storage apparatus **100** can be avoided, thereby improving the usage reliability of the energy storage apparatus **100**.

(49) The positive-electrode deformation member **40** is mounted in the first mating hole **203** and fixedly connected to a wall of the first mating hole **203**. The negative-electrode deformation member **50** is mounted in the second mating hole **204**, and is fixedly connected to a wall of the second mating hole **204**. The first avoidance groove **102** can avoid collision with the positive-electrode deformation member **40**, and the second avoidance groove **103** can avoid collision with the negative-electrode deformation member **50**. Exemplarily, the positive-electrode deformation member **40** may be fixedly connected to the wall of the first mating hole **203** by welding, so that the positive-electrode deformation member **40** is mounted in the first mating hole **203**. Alternatively or additionally, the negative-electrode deformation member **50** may be fixedly connected to the wall of the second mating hole **204** by welding, so that the negative-electrode deformation member **50** is mounted in the second mating hole **204**.
(50) As illustrated in FIGS. **4**, **6**, and **7**, FIG. **6** is a schematic exploded structural view of the positive-electrode unit **60** of the end cover assembly **120** illustrated in FIG. **3**, and FIG. **7** is a schematic sectional structural view of a first upper plastic member **61** of the positive-electrode unit **60** illustrated in FIG. **6**, taken along line II-II. In the embodiments of the disclosure, the upper plastic member may be implemented as at least one of the first upper plastic member **61** or a second upper plastic member **71**.
(51) The positive-electrode unit **60** includes the first upper plastic member **61**, a first conductive block **62**, a first terminal post **63**, a first sealing member **64**, and a first connector **65**. The first upper plastic member **61** is mounted to the top cover **20**. The first upper plastic member **61** includes a first mounting portion **611** and a first mating portion **612**. The first mating portion **612** is fixedly connected to a right side of the first mounting portion **611**. The first mounting portion **611** defines a first fitting groove **613**, a first avoidance hole **614**, and a first identification through-hole **615**. An opening of the first fitting groove **613** is defined on a right side surface (no reference number is illustrated) of the first mounting portion **611**. The first fitting groove **613** is recessed from the right side surface of the first mounting portion **611** towards a left side surface (no reference number is illustrated) of the first mounting portion **611**, that is, is recessed in the negative direction of the X-axis. The first fitting groove **613** has a first groove top surface **616** and a first groove bottom surface **617**. In the Z-axis direction, the first groove top surface **616** is spaced apart and is disposed opposite to the first groove bottom surface **617**.
(52) An opening of the first avoidance hole **614** is defined on a bottom surface (no reference number is illustrated) of the first mounting portion **611**. In an embodiment, the opening of the first avoidance hole **614** is positioned in the middle of the bottom surface of the first mounting portion **611**. The first avoidance hole **614** is recessed from the bottom surface of the first mounting portion **611** towards the first fitting groove **613**, that is, is recessed in the positive direction of the Z-axis as illustrated. The first avoidance hole **614** extends through the first groove bottom surface **617** of the first fitting groove **613**, and is in communication with the first fitting groove **613**. The first avoidance hole **614** is disposed opposite to the positive-electrode deformation member **40**. Exemplarily, the first avoidance hole **614** is a circular hole. In some other embodiments, the first avoidance hole **614** may also be a square hole or other irregular holes.
(53) An opening of the first identification through-hole **615** is defined on a top surface (no reference number is illustrated) of the first mounting portion **611**. In an embodiment, the opening of the first identification through-hole **615** is positioned in the middle of the top surface of the first mounting portion **611**. The first identification through-hole **615** is recessed from the top surface of the first mounting portion **611** towards the first fitting groove **613**, that is, is recessed in the negative direction of the Z-axis as illustrated. The first identification through-hole **615** extends through the first groove top surface **616** of the first fitting groove **613**, and is in communication with the first fitting groove **613**. The first identification through-hole **615** is of a cross-shape to identify that the polarity of the positive-electrode unit **60** is positive. In some other embodiments, the first identification through-hole **615** may also be of other shapes, as long as it can identify that

the polarity of the positive-electrode unit **60** is positive.
(54) In addition, the first mounting portion **611** includes an identification portion **611***a*. The identification portion **611***a* is positioned at an end of the first mounting portion **611** facing away from the first mating portion **612** to identify the polarity of the positive-electrode unit **60**. The identification portion **611***a* may be a chamfer. During assembly of the end cover assembly **120**, an operator or an intelligent device can quickly identify the first upper plastic member **61** of the positive-electrode unit **60** according to the identification portion **611***a* to distinguish the first upper plastic member **61** from the second upper plastic member **71** of the negative-electrode unit **70**, thereby improving assembly efficiency. In other words, the identification portion **611***a* may serve as a fool proof structure of the first upper plastic member **61**, preventing an operator from confusing the first upper plastic member **61** of the positive-electrode unit **60** and the second upper plastic member **71** of the negative-electrode unit **70**.
(55) The first mating portion **612** defines a first mounting groove **618** and a first through hole **619**. An opening of the first mounting groove **618** is defined on a top surface (no reference number is illustrated) of the first mating portion **612**. The first mounting groove **618** is recessed from the top surface of the first mating portion **612** towards a bottom surface (no reference number is illustrated) of the first mating portion **612**, that is, is recessed in the negative direction of the Z-axis as illustrated. The first mounting groove **618** is in communication with the first fitting groove **613**. The first mounting groove **618** has a groove bottom surface (no reference number is illustrated).
(56) The first through hole **619** extends through the first mating portion **612** in a thickness direction of the first mating portion **612**. In an embodiment, an opening of the first through hole **619** is defined on the bottom surface of the first mating portion **612**. The first through hole **619** is recessed from the bottom surface of the first mating portion **612** towards the first mounting groove **618**, that is, is recessed in the positive direction of the Z-axis as illustrated. The first through hole **619** extends through the groove bottom surface of the first mounting groove **618**, and is in communication with the first mounting groove **618**. The first through hole **619** is in communication with the first mounting hole **205** of the top cover **20**. Exemplarily, the first through hole **619** is a circular hole. In some other embodiments, the first through hole **619** may also be a square hole or other irregular holes.
(57) FIG. **8** is a schematic structural view of the first conductive block **62** of the positive-electrode unit **60** illustrated in FIG. **6**, viewed from another direction.
(58) The first conductive block **62** is mounted to the first upper plastic member **61**. The first conductive block **62** may be made of aluminum. In this embodiment, the first conductive block **62** includes a first extension portion **621** and a first welding portion **622**. The first welding portion **622** is fixedly connected to a right side of the first extension portion **621**. The first extension portion **621** includes a first top surface **623** and a first bottom surface **624** opposite to the first top surface **623**. Exemplarily, both the first top surface **623** and the first bottom surface **624** are flat surfaces.
(59) In an embodiment, the first extension portion **621** is mounted in the first fitting groove **613** and covers the first avoidance hole **614** of the first upper plastic member **61** and the first identification through-hole **615** of the first upper plastic member **61**. The first extension portion **621** is in interference fit with the first fitting groove **613**. The first top surface **623** of the first extension portion **621** abuts against the first groove top surface **616** of the first fitting groove **613** and covers the first identification through-hole **615** of the first upper plastic member **61**. The first bottom surface **624** of the first extension portion **621** abuts against the first groove bottom surface **617** of the first fitting groove **613** and covers the first avoidance hole **614** of the first upper plastic member **61**.
(60) In this embodiment, a color of the first extension portion **621** is different from a color of the first mounting portion **611**. For example, the first extension portion **621** is in a bright silver color, and the first mounting portion **611** is in a black color. The first top surface **623** of the first extension portion **621** exposes through the first identification through-hole **615**, so that a pronounced color

contrast is created to form a positive polarity symbol "+", thereby identifying the polarity of the positive-electrode unit **60**. Furthermore, the first extension portion **621** is in interference fit with the first fitting groove **613**, thus the first top surface **623** of the first extension portion **621** is in close contact with the first groove top surface **616** of the first fitting groove **613** without a gap. This prevents the first identification through-hole **615** from being suspended to form a shadow on the first top surface **623** of the first extension portion **621**, thereby improving legibility of the polarity identification.

(61) A thickness of the first welding portion **622** is greater than a thickness of the first extension portion **621**. The first welding portion **622** includes a second top surface **625** and a second bottom surface **626** opposite the second top surface **625**. The second bottom surface **626** of the first welding portion **622** is flush with the first bottom surface **624** of the first extension portion **621**. The second top surface **625** of the first welding portion **622** protrudes relative to the first top surface **623** of the first extension portion **621**.

(62) The first welding portion **622** defines a first fitting hole **627**. The first fitting hole **627** is positioned in the middle of the first welding portion **622**. The first fitting hole **627** extends through the first welding portion **622** in a thickness direction of the first welding portion **622** (i.e., the Z-axis direction as illustrated). Exemplarily, the first fitting hole **627** is a circular hole, and in other embodiments, the first fitting hole **627** may also be a square hole or other irregular holes.

(63) In addition, the first welding portion **622** is provided with a first protruding ring **622*a***. The first protruding ring **622*a*** is disposed on the first welding portion **622** and surrounds an inner wall of the first fitting hole **627**. In an embodiment, the first protruding ring **622*a*** is disposed on the second bottom surface **626** of the first welding portion **622** and protrudes from the second bottom surface **626** of the first welding portion **622** in a direction facing away from the second top surface **625**. A bottom surface of the first protruding ring **622*a*** is a flat surface. Exemplarily, the first protruding ring **622*a*** is in an annular shape and is coaxial with the first fitting hole **627**. In some other embodiments, the first protruding ring **622*a*** may also be in a rectangular ring shape or other irregular ring shapes.

(64) In this embodiment, a ratio of a thickness w of the first protruding ring **622*a*** to a height h of the first protruding ring **622*a*** ranges from 0.2 to 0.8. This can ensure that the first protruding ring **622*a*** has enough rigidity, so that the first protruding ring **622*a*** can abut against the first upper plastic member **61**, thereby tightly pressing the first sealing member **64**. In an embodiment, the height h of the first protruding ring **622*a*** ranges from 0.1 mm to 1.2 mm to ensure that the first protruding ring **622*a*** is sufficiently tall to effectively abut against the first upper plastic member **61**. The thickness w of the first protruding ring **622*a*** ranges 0.2 mm to 2.4 mm to ensure that the first protruding ring **622*a*** has a sufficient region in contact with the first upper plastic member **61**, thereby allowing a more uniform abutment against the first upper plastic member **61**.

(65) In an embodiment, the first welding portion **622** is disposed in the first mounting groove **618**. The first protruding ring **622*a*** abuts against the first upper plastic member **61**. The first fitting hole **627** of the first welding portion **622** is in communication with the first through hole **619** of the first upper plastic member **61**. The first protruding ring **622*a*** of the first welding portion **622** abuts against a groove bottom surface of the first mounting groove **618**. The second top surface **625** of the first welding portion **622** protrudes relative to the top surface of the first mounting portion **611** of the first upper plastic member **61**. A peripheral surface of the first welding portion **622** is spaced apart from a surface of the first mounting portion **611** facing towards the first mating portion **612** (i.e., the right side surface of the first mounting portion **611**). It may be understood that, the second top surface **625** of the first welding portion **622** protrudes relative to the top surface of the first mounting portion **611** of the first upper plastic member **61**. During welding of a connecting sheet to the second top surface **625** of the first welding portion **622**, it can not only prevent the first mounting portion **611** of the first upper plastic member **61** from being scratched by the connecting sheet so that the usage reliability of the energy storage apparatus **100** can be ensured, but also

prevent a busbar of the energy storage apparatus **100** from warping during connection of the busbar to a top surface of the first welding portion **622**, where warping of the busbar may result in a decrease in electrical connection reliability.
(66) In this embodiment, the first welding portion **622** includes a first fitting portion **628** and a first connecting portion **629**. The first connecting portion **629** is disposed on a top side of the first fitting portion **628**. The first connecting portion **629** extends outwards relative to a peripheral surface of the first fitting portion **628** to form a protruding ring. A bottom surface of the first fitting portion **628** serves as the second bottom surface **626** of the first welding portion **622**. A top surface of the first connecting portion **629** serves as the second top surface **625** of the first welding portion **622**. In addition, the first fitting hole **627** extends through the first fitting portion **628** and the first connecting portion **629**. The first protruding ring **622***a* is disposed on a surface of the first fitting portion **628** facing away from the first connecting portion **629** (i.e., the bottom surface of the first fitting portion **628**).
(67) In an embodiment, the first fitting portion **628** is disposed in the first mounting groove **618**. The first connecting portion **629** protrudes relative to the first mounting groove **618** and abuts against the top surface of the first mating portion **612** of the first upper plastic member **61**. A peripheral surface of the first connecting portion **629** is spaced apart from and opposite to the right side surface of the first mounting portion **611** of the first upper plastic member **61**. It may be understood that a gap between the right side surface of the first mounting portion **611** of the first upper plastic member **61** and the peripheral surface of the first connecting portion **629** may serve as an error accommodating region. When the first upper plastic member **61** shrinks due to thermal contraction, the first upper plastic member **61** and the first conductive block **62** can still be assembled in place, thereby ensuring the assembling reliability between the first upper plastic member **61** and the first conductive block **62**. In some other examples, the first connecting portion **629** may also be flush with the opening of the first mounting groove **618**, or the first connecting portion **629** may be recessed relative to the first mounting groove **618**.
(68) In addition, the peripheral surface of the first fitting portion **628** is covered by a side wall of the first mounting groove **618**. That is, the first mating portion **612** of the first upper plastic member **61** covers the peripheral surface of the first fitting portion **628**. A distance between the first connecting portion **629** and the top cover **20** improves the insulation performance between the first connecting portion **629** and the top cover **20**, and increases a creep age distance between the first conductive block **62** and the top cover **20**, thereby avoiding an arc short circuit between the first conductive block **62** and the top cover **20**.
(69) The first terminal post **63** includes a first flange portion **631** and a first post portion **632**. The first post portion **632** is disposed on a top side of the first flange portion **631**. In an embodiment, the first post portion **632** extends through the third mounting hole **104** of the lower plastic member **10**, the first mounting hole **205** of the top cover **20**, the first through hole **619** of the first upper plastic member **61**, the first protruding ring **622***a* of the first conductive block **62**, and the first fitting hole **627** of the first conductive block **62**, and is fixedly connected to the first conductive block **62**. The first post portion **632** is fixedly connected to a wall of the first fitting hole **627**. Exemplarily, the first post portion **632** may be fixedly connected to the wall of the first fitting hole **627** of the first conductive block **62** by welding. In addition, the first flange portion **631** may be positioned inside the lower plastic member **10**, and a top surface of the first flange portion **631** may abut against the lower plastic member **10**.
(70) The first sealing member **64** includes a first sealing portion **641** and a first secondary sealing portion **642**. The first secondary sealing portion **642** is disposed at a bottom of the first sealing portion **641** and surrounds the first sealing portion **641**. In an embodiment, the first sealing member **64** is sleeved on the first post portion **632** of the first terminal post **63** and is clamped between the first post portion **632** and a wall of the first mounting hole **205** of the top cover **20**. A top surface (no reference number is illustrated) of the first sealing member **64** abuts against a bottom surface of

the first upper plastic member **61**. A bottom surface (no reference number is illustrated) of the first sealing member **64** abuts against the top surface of the first flange portion **631** of the first terminal post **63**. The first sealing portion **641** of the first sealing member **64** is clamped between the first post portion **632** of the first terminal post **63** and the wall of the first mounting hole **205** of the top cover **20**. A surface of the first sealing portion **641** facing away from the first secondary sealing portion **642** (i.e., a top surface of the first sealing portion **641**) abuts against the bottom surface of the first upper plastic member **61**. A surface of the first secondary sealing portion **642** facing away from the first sealing portion **641** (i.e., a bottom surface of the first secondary sealing portion **642**) abuts against the top surface of the first flange portion **631** of the first terminal post **63**.
(71) It may be noted that, the first sealing member **64** not only can ensure stability of assembly between the first terminal post **63** and the lower plastic member **10** and stability of assembly between the first terminal post **63** and the top cover **20**, but also can prevent the first terminal post **63** from being in direct electrical contact with the top cover **20**, thereby realizing insulation between the first terminal post **63** and the top cover **20**.
(72) In the embodiments of the disclosure, the first protruding ring **622***a* of the first conductive block **62** abuts against the first mating portion **612** of the first upper plastic member **61**, so that the first mating portion **612** of the first upper plastic member **61** abuts against the top surface of the first sealing member **64**, and the first flange portion **631** of the first terminal post **63** abuts against the bottom surface of the first sealing member **64**. In this way, the first sealing member **64** is clamped between the first upper plastic member **61** and the first flange portion **631** of the first terminal post **63**, thereby ensuring the assembling stability of the first sealing member **64**, improving the sealing performance between the first terminal post **63** and the top cover **20**, ensuring the sealing performance of the energy storage apparatus **100**, and improving the assembly yield of the energy storage apparatus **100**.
(73) The first connector **65** is mounted at an inner side of the lower plastic member **10** and positioned at a side of the first flange portion **631** of the first terminal post **63** facing away from the first post portion **632**. In an embodiment, one end of the first connector **65** is electrically connected to the first flange portion **631** of the first terminal post **63**, and the other end of the first connector **65** is electrically connected to a positive tab of the jelly-roll. Exemplarily, the first connector **65** may be electrically connected to the first flange portion **631** of the first terminal post **63** and/or the positive tab of the jelly-roll by welding.
(74) As illustrated in FIGS. **4** and **9**, FIG. **9** is a schematic exploded structural view of the negative-electrode unit **70** of the end cover assembly **120** illustrated in FIG. **3**.
(75) The negative-electrode unit **70** includes a second upper plastic member **71**, a second conductive block **72**, a second terminal post **73**, a second sealing member **74**, and a second connector **75**. For a structure of each component in the negative-electrode unit **70**, an assembly relationship between a component and another component, an assembly relationship between a component and the lower plastic member **10**, and an assembly relationship between a component and the top cover **20**, reference may be made to the foregoing relevant illustration of the positive-electrode unit **60**, and details are not repeatedly described herein. In the embodiments of the disclosure, the conductive block may be implemented as at least one of the first conductive block **62** or the second conductive block **72**. In the embodiments of the disclosure, the terminal post may be implemented as at least one of the first terminal post **63** or the second terminal post **73**. In the embodiments of the disclosure, the sealing member may be implemented as at least one of the first sealing member **64** or the second sealing member **74**.
(76) The second upper plastic member **71** is mounted at a top side of the top cover **20**, is positioned at a left side of the first upper plastic member **61**, and is spaced apart from the first upper plastic member **61**. The second upper plastic member **71** includes a second mounting portion **711** and a second mating portion **712**. The second mating portion **712** is fixedly connected to a left side of the second mounting portion **711**. The second mounting portion **711** defines a second fitting groove

713, a second avoidance hole **714**, and a second identification through-hole **715**. The second mating portion **712** defines a second mounting groove **718** and a second through hole **719**. The second through hole **719** is in communication with the second mounting hole **206** of the top cover **20**. The second fitting groove **713** has a second groove top surface **716** and a second groove bottom surface **717**. The second avoidance hole **714** is disposed opposite to the negative-electrode deformation member **50**. The second identification through-hole **715** is in a rectangular shape to identify that the polarity of the negative-electrode unit **70** is negative. In some other embodiments, the second identification through-hole **715** may also of other shapes, as long as the second identification through-hole **715** can identify that the polarity of the negative-electrode unit **70** is negative. In the embodiments of the disclosure, the mounting portion may be implemented as at least one of the first mounting portion **611** or the second mounting portion **711**. In the embodiments of the disclosure, the mating portion may be implemented as at least one of the first mating portion **612** or the second mating portion **712**. In the embodiments of the disclosure, the fitting groove may be implemented as at least one of the first fitting groove **613** or the second fitting groove **713**. In the embodiments of the disclosure, the identification through-hole may be implemented as at least one of the first identification through-hole **615** or the second identification through-hole **715**. In the embodiments of the disclosure, the through hole may be implemented as at least one of the first through hole **619** or the second through hole **719**. In the embodiments of the disclosure, the mounting groove may be implemented as at least one of the first mounting groove **618** or the second mounting groove **718**. In the embodiments of the disclosure, the groove top surface may be implemented as at least one of the first groove top surface **616** or the second groove top surface **716**. In the embodiments of the disclosure, the groove bottom surface may be implemented as at least one of the first groove bottom surface **617** or the second groove bottom surface **717**.

(77) It may be noted that, in some other embodiments, the second mounting portion **711** may further include an identification portion (not illustrated). The identification portion is configured to identify the polarity of the negative-electrode unit **70**. The identification portion of the second mounting portion **711** is different from the identification portion **611***a* of the first mounting portion **611**, so that an operator may distinguish the second upper plastic member **71** of the negative-electrode unit **70** from the first upper plastic member **61** of the positive-electrode unit **60**.

(78) The second conductive block **72** is mounted to the second upper plastic member **71**. In this embodiment, the second conductive block **72** includes a second extension portion **721** and a second welding portion **722**. The second welding portion **722** is fixedly connected to a left side of the second extension portion **721**. The second welding portion **722** is provided with a second protruding ring **722***a*. The second protruding ring **722***a* is disposed on the second welding portion **722** and surrounds an inner wall of the second fitting hole **727**. The second protruding ring **722***a* is disposed on a fourth bottom surface **726** of the second welding portion **722**. The second protruding ring **722***a* protrudes from the fourth bottom surface **726** of the second welding portion **722** towards a direction facing away from a fourth top surface **725**. In the embodiments of the disclosure, the fitting hole may be implemented as at least one of the first fitting hole **627** or the second fitting hole **727**. In the embodiments of the disclosure, the protruding ring may be implemented as at least one of the first protruding ring **622***a* or the second protruding ring **722***a*. In the embodiments of the disclosure, the extension portion may be implemented as at least one of the first extension portion **621** or the second extension portion **721**. In the embodiments of the disclosure, the welding portion may be implemented as at least one of the first welding portion **622** or the second welding portion **722**.

(79) In an embodiment, the second extension portion **721** is mounted in the second fitting groove **713**. The second welding portion **722** is disposed in the second mounting groove **718**. The second fitting hole **727** of the second welding portion **722** is in communication with the second through hole **719** of the second mating portion **712**. The second protruding ring **722***a* of the second welding portion **722** abuts against a groove bottom surface of the second mounting groove **718**. The second

extension portion **721** is in interference fit with the second fitting groove **713**. A third top surface **723** of the second extension portion **721** abuts against the second groove top surface **716** of the second fitting groove **713**, and covers the second identification through-hole **715** of the second upper plastic member **71**. A third bottom surface **724** of the second extension portion **721** abuts against the second groove bottom surface **717** of the second fitting groove **713**, and covers the second avoidance hole **714** of the second upper plastic member **71**.

(80) In this embodiment, a color of the second extension portion **721** is different from a color of the second mounting portion **711**. For example, the second extension portion **721** is in a bright silver color, and the second mounting portion **711** is in a black color. The third top surface **723** of the second extension portion **721** exposes through the second identification through-hole **715**, so that a pronounced color contrast is created to form a negative polarity symbol "−", thereby identifying the polarity of the negative-electrode unit **70**. In addition, when the gas pressure inside the energy storage apparatus **100** is excessively high, both the positive-electrode deformation member **40** and the negative-electrode deformation member **50** are driven to flip by the gas pressure. In this way, the positive-electrode deformation member **40** extends through the first avoidance hole **614** and comes into contact with the first welding portion **622** so that the top cover **20** is electrically connected to the first conductive block **62**, and the negative-electrode deformation member **50** extends through the second avoidance hole **714** and comes into contact with the second welding portion **722** so that the top cover **20** is electrically connected to the second conductive block **72**. As a result, the positive-electrode unit **60** and the negative-electrode unit **70** are conducted to generate a short circuit, rendering the energy storage apparatus **100** unable to operate properly. Therefore, the gas pressure inside the energy storage apparatus **100** cannot be further increased, thereby improving the usage reliability of the energy storage apparatus **100**.

(81) In addition, the second welding portion **722** includes a second fitting portion **728** and a second connecting portion **729**. The first connecting portion **729** is disposed on a top side of the second fitting portion **728**. The second connecting portion **729** extends outwards relative to a peripheral surface of the second fitting portion **728** to form a protruding ring. In an embodiment, the second fitting portion **728** is mounted in the second mounting groove **718**, and the second connecting portion **729** protrudes relative to the second mounting groove **718** and abuts against a top surface of the second mating portion **712** of the second upper plastic member **71**. The peripheral surface of the second fitting portion **728** is covered by a side wall of the second mounting groove **718**. That is, the second mating portion **712** of the second upper plastic member **71** covers the peripheral surface of the second fitting portion **728**. A distance between the second connecting portion **729** and the top cover **20** improves the insulation performance between the second connecting portion **729** and the top cover **20**, and increases a creep age distance between the second conductive block **72** and the top cover **20**, thereby avoiding an arc short circuit between the second conductive block **72** and the top cover **20**. In the embodiments of the disclosure, the fitting portion may be implemented as at least one of the first fitting portion **628** or the second fitting portion **728**. In the embodiments of the disclosure, the connecting portion may be implemented as at least one of the first connecting portion **629** or the second connecting portion **729**.

(82) The second terminal post **73** includes a second flange portion **731** and a second post portion **732**. The second post portion **732** is fixedly connected to a top side of the second flange portion **731**. In an embodiment, the second post portion **732** extends through the fourth mounting hole **105** of the lower plastic member **10**, the second mounting hole **206** of the top cover **20**, the second through hole **719** of the second upper plastic member **71**, the second protruding ring **722*a*** of the second conductive block **72**, and the second fitting hole **727** of the second conductive block **72**. The second post portion **732** is fixedly connected to a wall of the second fitting hole **727** of the second conductive block **72**. In the embodiments of the disclosure, the flange portion may be implemented as at least one of the first flange portion **631** or the second flange portion **731**. In the embodiments of the disclosure, the post portion may be implemented as at least one of the first post

portion **632** or the second post portion **732**.
(83) The second sealing member **74** includes a second sealing portion **741** and a second secondary sealing portion **742**. The second secondary sealing portion **742** is fixedly connected to a bottom of the second sealing portion **741** and surrounds the second sealing portion **741**. In an embodiment, a top surface (no reference number is illustrated) of the second sealing portion **741** abuts against a bottom surface of the second upper plastic member **71**. A bottom surface (no reference number is illustrated) of the second sealing portion **74** abuts against a top surface of the second flange portion **731** of the second terminal post **73**. Part of the second sealing portion **741** is clamped between the second post portion **732** of the second terminal post **73** and a wall of the second mounting hole **206** of the top cover **20**. Part of the second sealing portion **741** is clamped between the second post portion **732** of the second terminal post **73** and a wall of the fourth mounting hole **105** of the lower plastic member **10**. In the embodiments of the disclosure, the sealing portion may be implemented as at least one of the first sealing portion **641** or the second sealing portion **741**. In the embodiments of the disclosure, the secondary sealing portion may be implemented as at least one of the first secondary sealing portion **642** or the second secondary sealing portion **742**.
(84) The second connector **75** is mounted at an inner side of the lower plastic member **10** and positioned at a side of the second flange portion **731** of the second terminal post **73** facing away from the second post portion **732**. In an embodiment, one end of the second connector **75** is electrically connected to the second flange portion **731** of the second terminal post **73**, and the other end of the second connector **75** is electrically connected to a negative tab of the jelly-roll. Exemplarily, the second connector **75** may be electrically connected to the second flange portion **731** of the second terminal post **73** and/or the positive tab of the jelly-roll by welding.
(85) In the embodiments of the disclosure, the protruding ring of the conductive block abuts against the upper plastic member, so that the upper plastic member abuts against the top surface of the sealing member. The flange portion of the terminal post abuts against the bottom surface of the sealing member. As such, the sealing member is clamped between the upper plastic member and the terminal post. In this way, the assembly stability of the sealing member is ensured, the sealing performance between the terminal post and the top cover is enhanced, and the sealing performance of the jelly-roll is ensured.
(86) It may be understood that, the energy storage apparatus **100** may also be an apparatus that has a function of storing power, such as a battery module, a battery pack, or a battery system. For example, the energy storage apparatus **100** is a battery pack. The energy storage apparatus **100** includes multiple jelly-rolls and multiple connecting sheets. Each connecting sheet is electrically connected between two jelly-rolls. The jelly-roll may have structures illustrated in the energy storage apparatus **100** described in the above embodiments. In this case, the multiple jelly-rolls may be connected in series, one end of each connection sheet is electrically connected to a positive electrode assembly of one jelly-roll, and another end of each connection sheet is electrically connected to a negative electrode assembly of another jelly-roll. Alternatively, the multiple jelly-rolls may also be connected in parallel, part of the connecting sheets is electrically connected between the positive electrode assemblies of the two jelly-rolls, and part of the connecting sheets is electrically connected between the negative electrode assemblies of the two jelly-rolls. Exemplarily, the connecting sheet may be an aluminum bar. It may be noted that, part of the jelly-rolls may also be arranged in series, and part of the jelly-rolls may be arranged in parallel. The embodiments of the disclosure do not specifically limit a connection manner of the multiple jelly-rolls in the battery pack.
(87) An electricity-consumption device is further provided in the disclosure. The electricity-consumption device includes the energy storage apparatus **100**. The energy storage apparatus **100** is configured to power the electricity-consumption device. The electricity-consumption device may be a new energy vehicle (e.g., hybrid or electric vehicle), an electricity storage station, a server, and other devices requiring electricity.

(88) The above illustrations are merely specific embodiments of the disclosure, but are not intended to limit the scope of protection of the disclosure. Any variation or replacement readily figured out by a person skilled in the art within the technical scope disclosed in the disclosure shall belong to the scope of protection of the disclosure. Without conflict, the embodiments of the disclosure and the features in the embodiments can be combined with each other. Therefore, the scope of protection of the disclosure shall be subject to the scope of protection of the claims.

Claims

1. An end cover assembly applicable to an energy storage apparatus and comprising: a top cover, wherein the top cover defines a mounting hole, and the mounting hole extends through the top cover in a thickness direction of the top cover; an upper plastic member mounted on the top cover, wherein the upper plastic member comprises a mounting portion and a mating portion, the mating portion is fixedly connected to one side of the mounting portion, the mounting portion defines a fitting groove and an identification through-hole, the identification through-hole is in communication with the fitting groove, and the mating portion defines a through hole extending through the upper plastic member in a thickness direction of the mating portion and being in communication with the mounting hole; a conductive block mounted on the upper plastic member, wherein: the conductive block comprises an extension portion and a welding portion, the welding portion is fixedly connected to one side of the extension portion, the extension portion is mounted in the fitting groove, a top surface of the extension portion covers the identification through-hole, the welding portion defines a fitting hole, the fitting hole extends through the welding portion in a thickness direction of the welding portion and is in communication with the through hole, the welding portion comprises a protruding ring, the protruding ring is disposed on the welding portion and surrounds an inner wall of the fitting hole, and the protruding ring is in contact with and abuts against the upper plastic member after assembly; a terminal post comprising a flange portion and a post portion, wherein the post portion is disposed on a top side of the flange portion, and the post portion extends through the mounting hole, the through hole, and the fitting hole and is fixedly connected to the conductive block; and a sealing member sleeved on the post portion and clamped between the post portion and a wall of the mounting hole, wherein a top surface of the sealing member abuts against a bottom surface of the upper plastic member, and a bottom surface of the sealing member abuts against a top surface of the flange portion, wherein the protruding ring is coaxial with the fitting hole of the welding portion and abuts against a bottom surface of a mounting groove of the upper plastic member; and wherein the post portion of the terminal post extends through a mounting hole of the lower plastic member, the mounting hole of the top cover, the through hole of the mating portion of the upper plastic member, the protruding ring of the welding portion of the conductive block, and the fitting hole of the welding portion of the conductive block.
2. The end cover assembly of claim 1, wherein a ratio of a thickness of the protruding ring to a height of the protruding ring ranges from 0.2 to 0.8.
3. The end cover assembly of claim 1, wherein a height of the protruding ring ranges from 0.1 mm to 1.2 mm.
4. The end cover assembly of claim 1, wherein a thickness of the protruding ring ranges from 0.2 mm to 2.4 mm.
5. The end cover assembly of claim 1, wherein the sealing member comprises a sealing portion and a secondary sealing portion, wherein the secondary sealing portion is positioned on a bottom of the sealing portion and surrounds the sealing portion, and the sealing portion is sleeved on the post portion and clamped between the post portion and the wall of the mounting hole; wherein a surface of the sealing portion that faces away from the secondary sealing portion abuts against the bottom surface of the upper plastic member, and a surface of the secondary sealing portion that faces away

from the sealing portion abuts against the top surface of the flange portion.

6. The end cover assembly of claim 1, wherein the conductive block comprises a fitting portion and a connecting portion, wherein the connecting portion is disposed on a top side of the fitting portion and extends outwards relative to a peripheral surface of the fitting portion to form the protruding ring, and the upper plastic member covers the peripheral surface of the fitting portion.

7. The end cover assembly of claim 6, wherein the fitting portion is received in the mounting groove, the welding portion comprises the fitting portion and the connecting portion, and the welding portion is received in the mounting groove.

8. The end cover assembly of claim 7, wherein the fitting groove comprises a groove top surface and a groove bottom surface opposite the groove top surface, the top surface of the extension portion abuts against the groove top surface, and a bottom surface of the extension portion abuts against the groove bottom surface.

9. The end cover assembly of claim 1, wherein the extension portion is in interference fit with the fitting groove, and the top surface of the extension portion abuts against a groove top surface of the fitting groove.

10. The end cover assembly of claim 1, wherein the post portion is fixedly connected to the inner wall of the fitting hole of the welding portion of the conductive block by welding.

11. An energy storage apparatus, comprising a housing and an end cover assembly, the end cover assembly being mounted on a top side of the housing, wherein the end cover assembly comprises: a top cover, wherein the top cover defines a mounting hole, and the mounting hole extends through the top cover in a thickness direction of the top cover; an upper plastic member mounted on the top cover, wherein the upper plastic member comprises a mounting portion and a mating portion, the mating portion is fixedly connected to one side of the mounting portion, the mounting portion defines a fitting groove and an identification through-hole, the identification through-hole is in communication with the fitting groove, and the mating portion defines a through hole extending through the upper plastic member in a thickness direction of the mating portion and being in communication with the mounting hole; a conductive block mounted on the upper plastic member, wherein the conductive block comprises an extension portion and a welding portion, the welding portion is fixedly connected to one side of the extension portion, the extension portion is mounted in the fitting groove, a top surface of the extension portion covers the identification through-hole, the welding portion defines a fitting hole, the fitting hole extends through the welding portion in a thickness direction of the welding portion and is in communication with the through hole, the welding portion comprises a protruding ring, the protruding ring is disposed on the welding portion and surrounds an inner wall of the fitting hole, and the protruding ring is in contact with and abuts against the upper plastic member after assembly; a terminal post comprising a flange portion and a post portion, wherein the post portion is disposed on a top side of the flange portion, and the post portion extends through the mounting hole, the through hole, and the fitting hole and is fixedly connected to the conductive block; and a sealing member sleeved on the post portion and clamped between the post portion and a wall of the mounting hole, wherein a top surface of the sealing member abuts against a bottom surface of the upper plastic member, and a bottom surface of the sealing member abuts against a top surface of the flange portion, wherein the protruding ring is coaxial with the fitting hole of the welding portion and abuts against a bottom surface of the mounting groove of the upper plastic member; and wherein the post portion of the terminal post extends through a mounting hole of the lower plastic member, the mounting hole of the top cover, the through hole of the mating portion of the upper plastic member, the protruding ring of the welding portion of the conductive block, and the fitting hole of the welding portion of the conductive block.

12. The energy storage apparatus of claim 11, wherein a ratio of a thickness of the protruding ring to a height of the protruding ring ranges from 0.2 to 0.8.

13. The energy storage apparatus of claim 11, wherein a height of the protruding ring ranges from

0.1 mm to 1.2 mm.

14. The energy storage apparatus of claim 11, wherein a thickness of the protruding ring ranges from 0.2 mm to 2.4 mm.

15. The energy storage apparatus of claim 11, wherein the sealing member comprises a sealing portion and a secondary sealing portion, wherein the secondary sealing portion is positioned on a bottom of the sealing portion and surrounds the sealing portion, and the sealing portion is sleeved on the post portion and clamped between the post portion and the wall of the mounting hole; wherein a surface of the sealing portion that faces away from the secondary sealing portion abuts against the bottom surface of the upper plastic member, and a surface of the secondary sealing portion that faces away from the sealing portion abuts against the top surface of the flange portion.

16. The energy storage apparatus of claim 11, wherein the conductive block comprises a fitting portion and a connecting portion, wherein the connecting portion is disposed on a top side of the fitting portion and extends outwards relative to a peripheral surface of the fitting portion to form the protruding ring, and the upper plastic member covers the peripheral surface of the fitting portion.

17. The energy storage apparatus of claim 16, wherein the upper plastic member defines a mounting groove, and the fitting portion is received in the mounting groove.

18. The energy storage apparatus of claim 17, wherein the through hole is in communication with the mounting groove, the fitting hole extends through the fitting portion and the connecting portion, and the protruding ring is disposed on a surface of the fitting portion away from the connecting portion.

19. The energy storage apparatus of claim 17, wherein the welding portion comprises the fitting portion and the connecting portion, and the welding portion is received in the mounting groove.

20. An electricity-consumption device, comprising an energy storage apparatus, the energy storage apparatus being configured to power the electricity-consumption device and comprising a housing and an end cover assembly, the end cover assembly being mounted on a top side of the housing, wherein the end cover assembly comprises: a top cover, wherein the top cover defines a mounting hole, and the mounting hole extends through the top cover in a thickness direction of the top cover; an upper plastic member mounted on the top cover, wherein the upper plastic member comprises a mounting portion and a mating portion, the mating portion is fixedly connected to one side of the mounting portion, the mounting portion defines a fitting groove and an identification through-hole, the identification through-hole is in communication with the fitting groove, and the mating portion defines a through hole extending through the upper plastic member in a thickness direction of the mating portion and being in communication with the mounting hole; a conductive block mounted on the upper plastic member, wherein the conductive block comprises an extension portion and a welding portion, the welding portion is fixedly connected to one side of the extension portion, the extension portion is mounted in the fitting groove, a top surface of the extension portion covers the identification through-hole, the welding portion defines a fitting hole, the fitting hole extends through the welding portion in a thickness direction of the welding portion and is in communication with the through hole, the welding portion comprises a protruding ring, the protruding ring is disposed on the welding portion and surrounds an inner wall of the fitting hole, and the protruding ring abuts is in contact with and against the upper plastic member after assembly; a terminal post comprising a flange portion and a post portion, wherein the post portion is disposed on a top side of the flange portion, and the post portion extends through the mounting hole, the through hole, and the fitting hole and is fixedly connected to the conductive block; and a sealing member sleeved on the post portion and clamped between the post portion and a wall of the mounting hole, wherein a top surface of the sealing member abuts against a bottom surface of the upper plastic member, and a bottom surface of the sealing member abuts against a top surface of the flange portion, wherein the protruding ring is coaxial with the fitting hole of the welding portion and abuts against a bottom surface of the mounting groove of the upper plastic member;

and wherein the post portion of the terminal post extends through a mounting hole of the lower plastic member, the mounting hole of the top cover, the through hole of the mating portion of the upper plastic member, the protruding ring of the welding portion of the conductive block, and the fitting hole of the welding portion of the conductive block.